

Executive Decision Report

Retail Business Intelligence Capstone

Executive-ready analysis of revenue, customers, geography, risks and strategic actions

Prepared for Executive Management • 2026

Executive Summary

- Objective: convert retail transaction data into decision-ready business insights.
- Core KPIs: Total Revenue, Total Orders, Average Order Value (AOV), Total Quantity.
- Focus areas: revenue drivers, customer segments, geographic performance, temporal trends and business risks.
- Decision output: three evidence-based strategic recommendations with scenario-based ROI and a 0–6 month implementation roadmap.
- Important data limitation: CAC and Churn Rate cannot be directly calculated from the supplied dataset because required fields are absent.

Business & Analytical Objective

- Business question: What is driving revenue, where is performance concentrated, and which actions should management prioritize?
- Source fields include order_id, order_date, customer_segment, city, category, quantity, unit_price, discount_pct, payment_status, year, month, month_name, gross_sales, discount_amount and net_sales.
- Analytical period: use the period represented in the cleaned dataset.
- Decision principle: recommendations should be traceable to observed data rather than unsupported assumptions.

Methodology

1. Load and validate the cleaned dataset.
2. Calculate executive KPIs from net_sales, order_id and quantity.
3. Analyze revenue by category, customer segment and city.
4. Examine monthly revenue trends and payment/discount patterns.
5. Identify evidence-based risk areas.
6. Translate findings into three strategic recommendations.
7. Build scenario-based ROI projections and a phased implementation timeline.

Executive KPI Dashboard

- TOTAL REVENUE — Dataset-derived: `net_sales.sum()`

- TOTAL ORDERS — Dataset-derived: unique `order_id` count

- AVERAGE ORDER VALUE — Total Revenue ÷ Total Orders

- TOTAL QUANTITY — Dataset-derived: `quantity.sum()`

- AVERAGE DISCOUNT — Dataset-derived: mean `discount_pct`

- Use exact values from `Executive_Analysis.ipynb`; no unsupported figures are inserted here.

Key Revenue Drivers

- Primary analysis: Revenue by Category using net_sales.
- Measure each category's revenue contribution and share of total revenue.
- Identify the highest-contributing category and the next-largest categories from the dataset.
- Executive implication: distinguish between growth opportunities and concentration exposure.
- Recommended visual: horizontal bar chart of category revenue + revenue-share callouts.

Customer Segment Analysis

- Analyze net_sales by customer_segment.
- Calculate each segment's revenue contribution and share of total revenue.
- Compare the largest segments with the overall revenue mix.
- Executive implication: align retention, cross-sell and service priorities with the segments that materially contribute to revenue.
- Recommended visual: segment revenue bar chart with percentage-share labels.

Geographic & Temporal Analysis

- Geographic view: rank cities by net_sales and highlight the top revenue-producing locations.
- Temporal view: compare monthly net_sales to identify peaks, troughs and volatility.
- Use city-level analysis rather than inventing latitude/longitude where coordinates are not present.
- Executive implication: identify markets to protect, markets to develop and periods requiring operational attention.
- Recommended visuals: Top 15 Cities bar chart + Monthly Revenue Trend line/area chart.

Risk Areas

- Revenue concentration — assess whether a small number of categories or cities contribute a disproportionate share.
- Payment performance — quantify unsuccessful/pending payment activity if supported by payment_status.
- Discount exposure — examine revenue associated with higher discount levels and the average discount.
- Geographic concentration — assess dependency on a small set of cities.
- Revenue volatility — use the monthly trend to identify unusually high or low periods.
- CAC and Churn — not directly measurable from this dataset because acquisition-cost and customer-activity/churn fields are absent.

Three Strategic Recommendations

- 01 • Protect and expand high-performing revenue drivers — use category and segment evidence to focus commercial effort where revenue contribution is strongest.
- 02 • Diversify concentrated exposure — develop secondary categories/cities where analysis shows material dependence on a small number of drivers.
- 03 • Strengthen performance controls — monitor payment outcomes, discount exposure and monthly revenue movement through recurring KPI reviews.
- Each recommendation should be finalized with the exact dataset finding, owner, KPI target and evidence before submission.

ROI & Implementation Roadmap

- ROI formula: $(\text{Expected Benefit} - \text{Investment}) \div \text{Investment} \times 100$.
- ROI figures must be labeled as scenario-based estimates when investment/benefit assumptions are not contained in the dataset.
- 0–30 days: validate findings, assign owners, establish KPI baselines.
- 31–90 days: implement selected initiatives and monitor revenue/operational KPIs.
- 3–6 months: measure realized impact, compare against scenario ROI, optimize and scale successful actions.
- Use an assumption table for each recommendation: Investment | Expected Benefit | Projected ROI.

Conclusion & Executive Action Plan

- Decision sequence: validate the strongest revenue drivers → confirm the most material risks → approve three focused initiatives.
- Track: Revenue, Orders, AOV, Quantity, category/segment mix, city mix, payment outcomes and discount behavior.
- Review cadence: 30 / 60 / 90 days, followed by a 3–6 month impact review.
- Final submission package: Executive Decision Report PDF + Final Analytical Jupyter Notebook + Dashboard Link.
- Success criterion: every recommendation is traceable to a documented finding or clearly labeled scenario assumption.